

Sensor applications

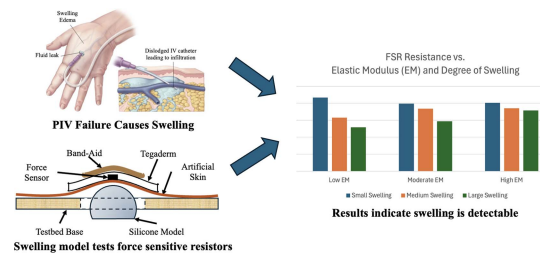
Force-Sensitive Resistors for Detecting Peripheral IV Failure

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Manuscript received 21 May 2025; revised 3 July 2025 and 7 August 2025; accepted 25 August 2025. Date of publication 9 September 2025; date of current version 15 September 2025.

Abstract—Peripheral intravenous catheters fail frequently, with some estimates of failure rates as high as 50% globally. In this letter, the potential for using force-sensitive resistors (FSRs) for the detection of swelling caused by infiltration or extravasation associated with peripheral IV failures (PIVIEs) is explored. Silicone models with elastic moduli and shape consistent with that experienced by body tissue during PIVIEs were integrated into a testbed with a layer of artificial skin, transparent film dressing, FSR, and bandage. The response of an FSR manufactured by Tekscan with a standard response range up to 1 lb (4.4 N) was measured for each combination of three swelling heights (8, 12, and 18 mm) and elastic moduli (~390, ~490, and ~540 kPa) and compared to the response of a precise force sensor (SingleTact 1.0 N) to determine the feasibility of using FSRs for detecting PIVIEs. Results show that the FSR resistance consistently decreases with increasing swelling height for a given elastic modulus. The coefficient of variability in the force-sensitive resistor output varies from 0.60% to 31.3% with the lowest variability occurring for the smallest amount of swelling. When filtering is introduced to remove outlying events in the FSR response, however, the range of variability drops to between 0.42% and 10.2%. While these experiments are preliminary, they nevertheless demonstrate the potential for using FSRs in single-use, disposable patches for monitoring IVs and detecting failure due to infiltration and swelling.



Index Terms—Sensor applications, extravasation, force-sensitive resistors, infiltration, intravenous.

I. INTRODUCTION

Peripheral intravenous (PIV) catheters are widely used to deliver fluids and medications directly into the bloodstream, with about 90% of hospitalized patients in the U.S. receiving one annually totaling over 300 million insertions in the U.S. [1] and over one billion worldwide [2]. Despite their ubiquity, PIVs frequently fail when the catheter dislodges from the vein, causing fluids to leak into surrounding tissue Fig. 1—a complication known as infiltration (nonvesicant) or extravasation (vesicant). These PIV infiltration and extravasation events (PIVIEs) are common, with failure rates between 35% and 50% in adults [3] often leading to extended hospital stays, increased costs, and worse patient outcomes [4]. Swelling is one of the earliest and most consistent signs of PIVIEs, yet current monitoring methods rely on intermittent visual checks and manual palpation, which are subjective and labor-intensive. Furthermore, existing technological solutions are often costly, invasive, or unsuitable for healthcare settings with limited resources. To address these constraints, this study explores the feasibility of using low-cost, small-footprint force-sensitive resistors (FSRs) to detect PIVIEs.

II. BACKGROUND

Several sensor technologies have been demonstrated for measuring PIVIEs, including bioimpedance, temperature, radio frequency,

optical, ultrasound, gamma scintillation, vein fluid pressure, and deformation/strain sensors. A thorough review of these sensor technologies can be found in [5]. Pressure, force, or strain sensors that measure swelling or deformation of the skin resulting from PIVIEs are of particular interest to portable “patch” style solutions and are sufficiently low in cost, power, and size to be used in a broad range of healthcare settings. Commercially available FSRs are one of the promising candidates among these types of sensors because of their high sensitivity, simple interface electronics, and low cost. These thin-film sensors typically consist of a matrix of both conductive and nonconductive particles sandwiched between conductive electrodes. As force is applied to the FSR, the conductive particles move closer together, causing the resistance of the sensor to decrease. Thru-mode configurations of FSRs provide more sensitivity and smaller dynamic range, while shunt-mode configurations provide broader dynamic range but reduced sensitivity [6]. Regardless of configuration, however, FSRs are generally less expensive and more physically flexible than other pressure and force sensors, which make them attractive for a wide range of wearable and low-cost applications. In a testament to their usefulness, FSRs have been demonstrated in biomedical and biomechanical applications, including characterizing gait [7], mapping finger forces to support robotics development [8], monitoring sleep-related breathing problems [9], [10], detecting Parkinsons and neurodegenerative diseases [11], [12], and diagnosing cardiovascular disease by externally monitoring blood flow [13].

Of particular interest to the detection of PIVIEs, however, are studies that have successfully demonstrated the use of FSRs in biomedically relevant instances of swelling. For example, the swelling of tissue in the

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Associate Editor: Jeong Bong (JB) Lee.

Digital Object Identifier 10.1109/LENS.2025.3605173

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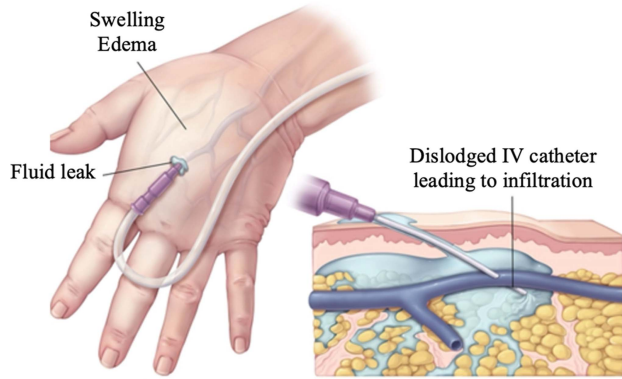


Fig. 1. Schematic of a PIVIE.

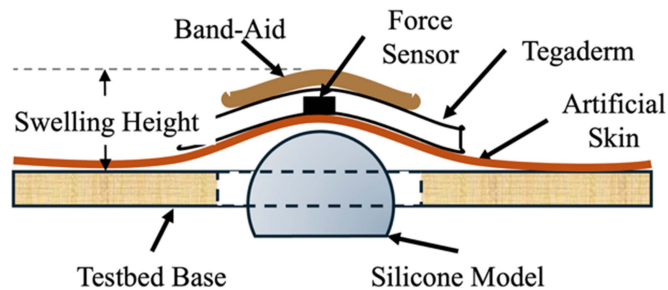


Fig. 2. "Force Sensor" for the swelling model used in this study was either the ESS102 FSR or the SingleTact reference sensor.

leg (i.e., edema) due to fluid retention caused by heart failure, kidney malfunction, or other conditions has been successfully monitored using a smart cuff that uses FSRs to gauge edema at multiple points along the cuff [14], [15]. This approach is suitable for both healthcare and home settings and compensates for movement that would otherwise interfere with mapping pressures of interest. FSRs have also been integrated into compression sleeves for the prevention of irreversible limb swelling caused by breast cancer and related surgeries [16].

While these studies are promising for monitoring swelling in PIVIEs, the benefits of FSRs also come with drawbacks. Among these drawbacks, repeatability (where the same input force produces significantly different output resistance) has been identified as a major issue that remains to be addressed [17]. Other limitations of FSRs include sensitivity to temperature, curvature, and movement [18], [19]. Applications that are at least partially immune to these drawbacks are especially attractive. The detection of PIVIEs promises to be one of these applications. An integrated "patch" style design using an FSR to detect swelling could be designed as a single-use and disposable sensor, typically active for no more than a week. In a vast majority of cases, the sensor would be used to detect only a single swelling event before IV and patch replacement, thereby making repeatability less relevant to detecting PIVIEs. Whether or not the remaining drawbacks and performance limits of the FSR preclude its use in detecting PIVIEs, however, is the focus of this feasibility study.

III. METHODS

To evaluate the potential of an FSR to detect swelling associated with PIVIEs, a model of swelling events was constructed (see Fig. 2). The model consists of five layers:

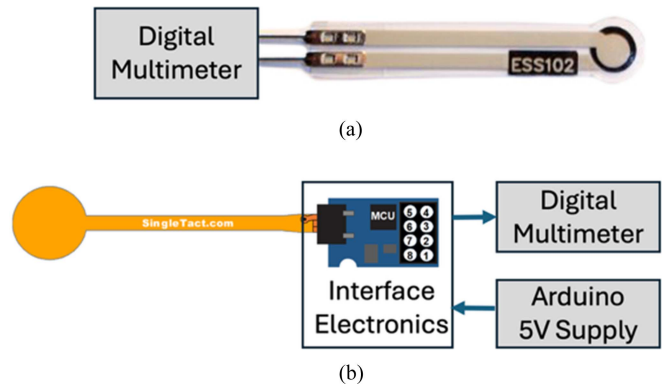


Fig. 3. Testing configuration.

- 1) a silicone rubber hemisphere which simulates uniform, circular swelling with elastic properties consistent with actual swelling events;
- 2) silicone-based artificial skin;
- 3) a reference sensor layer consisting of an FSR (Tekscan ESS102) or a reference sensor (SingleTact);
- 4) transparent film dressing (3M Tegaderm);
- 5) an adhesive bandage (Band-Aid tough strip).

The silicone model and overlying artificial skin form a model of a PIVIE and the resulting swelling. Tegaderm is commonly used for keeping PIVs in place, and the adhesive bandage creates a compressive force on the FSR formed by differences in elasticity of the model and the overlying bandage. After the Tegaderm was placed to hold the FSR in place, the location of the FSR center was marked on the testbed base to minimize variability in the placement of the silicone models between experiments.

Silicone models of three different elastic moduli (EM) were constructed for testing (~ 390 kPa, ~ 490 kPa, ~ 540 kPa) using three different silicone materials of varying Shore hardness values, with EM verified through stress-strain testing prior to experimental use, thereby serving as reasonable estimates of the swelling of human tissue and skin during PIVIEs. The EM of human skin varies based on individual characteristics and anatomical location but, on average, the skin EM is 100–200 kPa [20]. Edema, swelling, or inflammation can raise this by $2\times$ to over $7\times$ [21], [22]. In addition to varying EM, swelling models with heights of 8, 12, and 18 mm were constructed for each elastic modulus for a total of nine discrete PIVIE models. These heights are consistent with typical human edema/swelling values [23].

To eliminate potential sources of noise or systemic errors caused by interface electronics, the output of the ESS102 FSR was measured and recorded directly using a Fluke 77-IV digital multimeter [see Fig. 3(a)]. Raw data were collected every 1–1.5 s; small variations in sampling rate did not affect the validity of the results. Experimental data are presented for: 1) raw, unprocessed data; 2) data that were filtered to remove outliers; and 3) data that were filtered to remove events that were dominated by outliers. For 2), data collected from an entire experiment (corresponding to a particular swelling height and elastic modulus) were considered a single population, and outliers greater than two standard deviations from the mean were removed from the dataset. In order to consider time as a relevant parameter to detecting potentially anomalous data, in 3), the data were first filtered to flag sudden changes in the sensor output; subsequent events (data between these flags) were either retained (in the case where outliers were largely absent from an event) or removed (in the case where outliers were prevalent during an event).

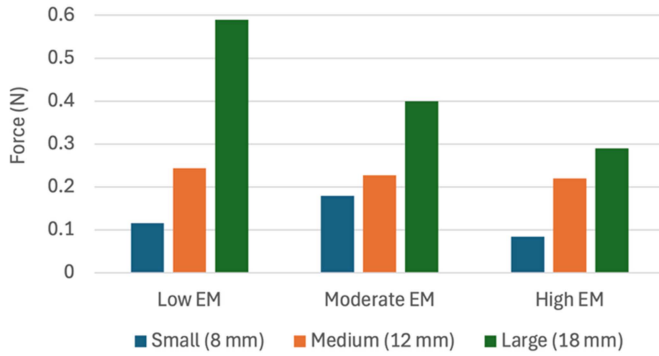


Fig. 4. Reference sensor response, i.e., Force (N), is calculated directly from manufacturer calibration data and indicates an expected increase in force applied to the sensor with increasing swelling height.

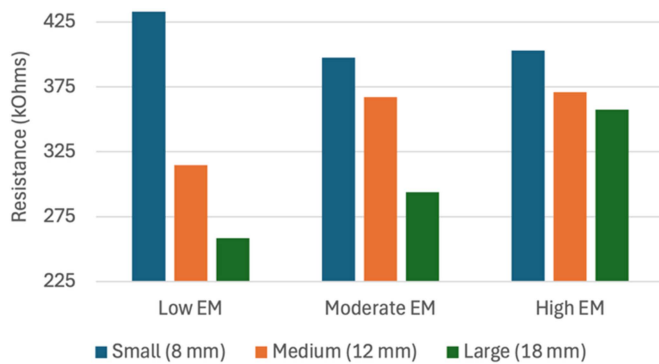


Fig. 5. Decrease in FSR resistance with increasing swelling height indicates a corresponding increase in force applied to the FSR.

A precise, calibrated force sensor of similar size to the FSR and manufactured by SingleTact (singletact.com) with a 1 N range was used as a reference to measure the actual force experienced by the FSR in the nine experimental conditions detailed above. The output of the reference sensor was measured as an analog voltage using interface electronics provided by the manufacturer that were uniquely paired to the SingleTact sensor to ensure the accuracy of calibration. Power was provided to the SingleTact interface electronics using the regulated 5 V output from an Arduino Uno, powered by USB. As with the ESS102 FSR, the output of the SingleTact sensor was collected and recorded manually using a Fluke 77-IV digital multimeter to minimize sources of systemic error or noise that were external to the sensor and sensing environment itself [see Fig. 3(b)].

IV. RESULTS AND DISCUSSION

The response of the reference sensor and the FSR to three different swelling conditions (small, medium, and large) and three different EM (low, moderate, and high) is shown in Figs. 4 and 5, respectively. As expected, the force experienced by the reference sensor increases and the FSR resistance decreases as the silicone model “pushes” upward against the overlying bandage.

Despite these similarities in overall response, the reference sensor and FSR exhibit some important differences. Since the FSR response is nonlinear and exhibits less sensitivity at higher forces, the response of the FSR at larger swelling heights is more muted than that of the reference sensor. Furthermore, neither the reference sensor nor the FSR exhibits a consistent trend in response across different EM. As EM

Table 1. Coefficient of Variability (%) in Sensor Output

		Swelling Height		
	EM	Small (8 mm)	Med (12 mm)	Large (18 mm)
FSR (Raw)	Low	2.42%	11.0%	31.3%
FSR (Outlier Filtered)		2.36%	5.14%*	12.7%*
FSR (Event Filtered)		2.40%	6.12%*	10.2%*
Reference (Raw)		6.47%	8.07%	7.45%
FSR (Raw)	High	0.60%	14.0%	6.32%
FSR (Outlier Filtered)		0.46%*	1.08%*	0.78%*
FSR (Event Filtered)		0.42%*	1.08%*	0.78%*
Reference (Raw)		2.89%	3.04%	1.11%

*F-test for equality of variance indicated that the variance of filtered data was significantly different than raw data ($p < 0.001$)

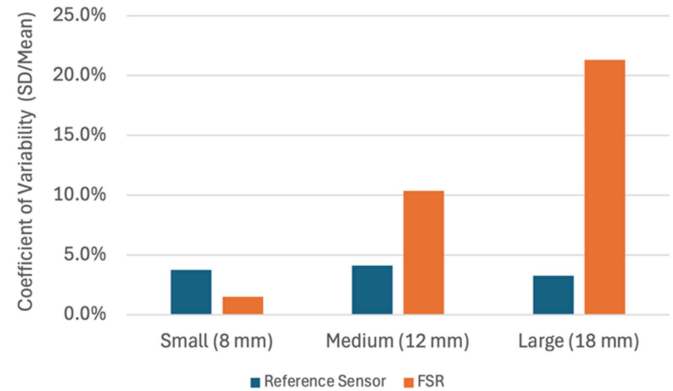


Fig. 6. Coefficient of variability increases with increasing swelling height for the FSR (raw data).

increases, one would expect that the silicone model is firmer and resists deformation more, thereby transferring more of the applied force to the sensor and increasing the sensor response. While this is true in some cases, most cases in Figs. 4 and 5 indicate that there is either no change or a decrease in response as EM increases for a given swelling height. This can likely be attributed to a second, competing (opposing) factor that influences sensor response, where higher EM (i.e., firmer) materials deform less for the same pressure. Less deformation leads to a sharper curvature and a “force gradient” toward the edges of the sensor, which, in turn, results in a lower average pressure across the sensor and a diminished sensor response. These curvature effects in FSRs have been confirmed in previous studies [18], [19].

In addition to these inconsistencies, the variability in FSR response over time (measured as the coefficient of variability and calculated as $100\% \times \text{standard deviation/mean}$) produces instability that, at swelling heights above 8 mm, is substantially greater than that of the reference sensor (see Fig. 6). Sudden resistance changes in force-sensitive resistors (FSRs) under static conditions are common and can be attributed to microstructural shifts within the sensor matrix itself, including rearrangement of conductive particles in the polymer matrix, which can create or break conductive pathways [24] and threshold-based polymer relaxation events, where internal stresses cause abrupt yielding or slip of the matrix [25]. In addition, minor shifts in pressure distribution or contact geometry—such as localized flattening or micro-slippage—can sharply alter sensor response, even when the overall load appears unchanged [26].

To address this variability, two filtering methods were used to process the raw data. The results were then compared to raw data using the coefficient of variability as a metric of comparison for two EM (low

and high) and three swelling heights: small (8 mm), medium (12 mm), and large (18 mm). The first filtering method (outlier filtered) removed data points greater than two standard deviations outside the mean. In the second filtering method (event filtered), the data were first filtered to flag any point at which a data point abruptly increased or decreased by more than half a standard deviation from its neighboring data point. These flags identified distinct regions separated by abrupt changes in the sensor output. If data points in between these distinct changes (i.e., events) were predominantly outliers, all the data in the event were removed. This approach effectively removed outlying events from the dataset. Both methods reduced variability in the data for all datasets, with no distinct advantage of one filtering method over the other (see Table 1).

V. CONCLUSION

The feasibility of using FSRs for detecting swelling associated with infiltration and extravasation in PIV failure has been studied. Results show that a single FSR can detect swelling as a monotonic decrease in resistance, regardless of the elasticity of the swelling event. Filtering that ignores temporal behavior, as well as filtering that considers FSR response over time, reduces the variability in FSR response to levels comparable to a precision force sensor. These results confirm the feasibility of using FSRs in portable, single-use PIV failure detection systems. However, the current study is limited by the fact that experiments were performed using laboratory-based models with a rigid base that does not fully replicate deeper tissue elasticity. An important topic for future work is to optimize sensor system design and test advanced prototypes on animal models and human tissue to confirm the generalizability of the findings in this study.

ACKNOWLEDGMENT

This letter was supported in part by the CoMotion Innovation Gap Fund at the University of Washington.

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